



A very interesting scientific publication

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Abstract. Hola

1 Shannon Explanation

Defined by Claude Shannon in 1984, Shannon's entropy has become one of the cornerstones of information theory. With applications in file compression (Huffman, 1952), DDoS attack detection (Lakhina et al., 2005) and more recently a key aspect of hallucination detection for LLMs (Farquhar et al., 2024). It is expressed as:

$$H(X) = - \sum_i p(x_i) \log(p(x_i)) \quad (1)$$

where $p_i(x_i)$ is the probability of the discrete event x_i to occur. It can be approximated to a measure of uniformity of the information inside a probabilistic distribution.

In the context of swarms of agents you can define the entropy not in terms of a discrete pool of events but connecting every term of the summatory into an agent with an internal distribution $p_i(t)$. Concretely, when considering every agent there are two possible expressions for the distribution corresponding with:

1. **Predictive Entropy:** Every LLM by selecting between the space of tokens creates a probability distribution based upon the likeliness of a token to correspond with the next best option, that is called logist. In the distribution comes the intuition of uncertainty, connecting the dispersion of it with a context in which multiple tokens could occupy the next space.
2. **Semantic Entropy:** As an improvement from predictive entropy in applications where the separation of syntax and semantics is crucial like hallucination detection (Farquhar et al., 2024). This kind clusters different tokens into semantic similar structures. That way, the distribution is normalized considering

the belonging to those structures and consequently the syntax uncertainty is removed from the distribution.

In this paper we will use the first definition of entropy, because the uncertainty in syntax will contain an important amount of information given when two different agents start interacting the style differences probably creates a considerable increase in entropy by adding much more context and tokens in to the context of each agent.

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2 Controller-Owned Board Storage Append-Only

Chainlink issue #16 establishes the storage boundary for the experiment's message board. The implementation uses a controller-owned SQLite database, not a database exposed to the Pi agent. The public storage interface is deliberately small:

```
with BoardStore.initialize(database_path,  
                             agent_workspace_roots=roots) as board:  
    record = board.append_message(run_id, agent_id, message)
```

The service generates the UTC server timestamp and UTF-8 byte count. The SQLite table contains the following fields:

Field	Constraint or meaning
sequence_id	INTEGER PRIMARY KEY AUTOINCREMENT
run_id	Required non-empty text
agent_id	Required non-empty text
server_timestamp	Required controller-generated UTC text
message_body	Required non-empty UTF-8 text
message_size	Required byte length of the UTF-8 body

Append-only behavior is enforced in the database itself. Triggers reject both UPDATE and DELETE, so the invariant

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does not depend only on callers using the Python interface. The low-level iterator orders rows by `sequence_id`; cursor semantics, C0/C1/C2 visibility, agent identity derivation, and Pi extension tools remain follow-up work.

The database path is checked against agent workspace roots, including paths that resolve through symlinks. The board service retains the database path and connection. Neither is passed to Pi, mounted into Bubblewrap, or made available to an agent task directory.

The implementation and regression tests were delivered in commit 264af09 and pushed to `origin/main` as part of the merge at d39bd82. Verification recorded for issue #16 is 39 passing unittest cases, successful runtime configuration validation, successful Python compilation checks, and a clean `git diff -check`. The tests cover schema creation, required-field and size constraints, UTF-8 sizing, monotonic ordering, symlink-aware workspace rejection, and direct SQL UPDATE/DELETE rejection. No subscription-backed model request was made.

References

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